CYBER ATTACK TRAFFIC CLASSIFICATION (CIC-IDS2017)

Input Feature Dictionary & Architectural Meaning: Complete specification of all 78 bidirectional flow telemetry features utilized by the classification model pipeline.

#	Feature Name (CICFlowMeter Attribute)	Technical Meaning & Network Security Context
1	`Destination Port`	Target destination transport layer port number (e.g., 80, 443, 21, 22).
2	`Flow Duration`	Total duration of the bidirectional network flow in microseconds (µs).
3	`Total Fwd Packets`	Total count of packets transmitted in the forward direction (source to dest).
4	`Total Backward Packets`	Total count of packets transmitted in the backward direction (dest to source).
5	`Total Length of Fwd Packets`	Cumulative payload length (in bytes) of all forward packets.
6	`Total Length of Bwd Packets`	Cumulative payload length (in bytes) of all backward packets.
7	`Fwd Packet Length Max`	Maximum payload size (in bytes) observed among forward packets.
8	`Fwd Packet Length Min`	Minimum payload size (in bytes) observed among forward packets.
9	`Fwd Packet Length Mean`	Arithmetic mean of forward packet payload lengths (in bytes).
10	`Fwd Packet Length Std`	Standard deviation of forward packet payload lengths.
11	`Bwd Packet Length Max`	Maximum payload size (in bytes) observed among backward packets.
12	`Bwd Packet Length Min`	Minimum payload size (in bytes) observed among backward packets.
13	`Bwd Packet Length Mean`	Arithmetic mean of backward packet payload lengths (in bytes).
14	`Bwd Packet Length Std`	Standard deviation of backward packet payload lengths.
15	`Flow Bytes/s`	Flow throughput rate measured as total bytes transmitted per second.
16	`Flow Packets/s`	Flow throughput rate measured as total packets transmitted per second.
17	`Flow IAT Mean`	Mean inter-arrival time (in µs) between consecutive packets in the flow.
18	`Flow IAT Std`	Standard deviation of packet inter-arrival times across the flow.
19	`Flow IAT Max`	Maximum inter-arrival time (in µs) between consecutive packets in the flow.
20	`Flow IAT Min`	Minimum inter-arrival time (in µs) between consecutive packets in the flow.
21	`Fwd IAT Total`	Total elapsed time (in µs) between the first and last packet in the forward direction.
22	`Fwd IAT Mean`	Mean inter-arrival time (in µs) between consecutive forward packets.
23	`Fwd IAT Std`	Standard deviation of inter-arrival times between forward packets.
24	`Fwd IAT Max`	Maximum inter-arrival time (in µs) observed between forward packets.
25	`Fwd IAT Min`	Minimum inter-arrival time (in µs) observed between forward packets.
26	`Bwd IAT Total`	Total elapsed time (in µs) between the first and last packet in the backward direction.
27	`Bwd IAT Mean`	Mean inter-arrival time (in µs) between consecutive backward packets.
28	`Bwd IAT Std`	Standard deviation of inter-arrival times between backward packets.
29	`Bwd IAT Max`	Maximum inter-arrival time (in µs) observed between backward packets.
30	`Bwd IAT Min`	Minimum inter-arrival time (in µs) observed between backward packets.
31	`Fwd PSH Flags`	Number of times the PSH (Push) TCP flag was asserted in forward packets.
32	`Bwd PSH Flags`	Number of times the PSH (Push) TCP flag was asserted in backward packets.
33	`Fwd URG Flags`	Number of times the URG (Urgent) TCP flag was asserted in forward packets.
34	`Bwd URG Flags`	Number of times the URG (Urgent) TCP flag was asserted in backward packets.
35	`Fwd Header Length`	Total header size (in bytes) for forward packets.
36	`Bwd Header Length`	Total header size (in bytes) for backward packets.
37	`Fwd Packets/s`	Transmission throughput of forward packets per second.

#	Feature Name (CICFlowMeter Attribute)	Technical Meaning & Network Security Context
38	`Bwd Packets/s`	Transmission throughput of backward packets per second.
39	`Min Packet Length`	Smallest packet payload size (in bytes) observed in the entire flow.
40	`Max Packet Length`	Largest packet payload size (in bytes) observed in the entire flow.
41	`Packet Length Mean`	Mean payload size (in bytes) of all packets transmitted across the flow.
42	`Packet Length Std`	Standard deviation of packet payload sizes across the flow.
43	`Packet Length Variance`	Statistical variance of packet payload sizes across the flow.
44	`FIN Flag Count`	Count of packets with the FIN (Finish connection) TCP control flag set.
45	`SYN Flag Count`	Count of packets with the SYN (Synchronize sequence numbers) flag set.
46	`RST Flag Count`	Count of packets with the RST (Reset connection) TCP flag set.
47	`PSH Flag Count`	Total count of packets across both directions with the PSH flag set.
48	`ACK Flag Count`	Count of packets with the ACK (Acknowledgment) TCP flag set.
49	`URG Flag Count`	Total count of packets across both directions with the URG flag set.
50	`CWE Flag Count`	Count of packets with the CWE (Congestion Window Reduced) flag set.
51	`ECE Flag Count`	Count of packets with the ECE (ECN-Echo) congestion notification flag set.
52	`Down/Up Ratio`	Download/Upload ratio (ratio of backward packets to forward packets).
53	`Average Packet Size`	Average size of packets observed across the bidirectional flow.
54	`Avg Fwd Segment Size`	Average segment size transmitted in the forward direction.
55	`Avg Bwd Segment Size`	Average segment size transmitted in the backward direction.
56	`Fwd Header Length.1`	Duplicate header length metric recorded for forward packets.
57	`Fwd Avg Bytes/Bulk`	Average number of bytes transmitted per bulk transmission in forward flow.
58	`Fwd Avg Packets/Bulk`	Average number of packets per bulk transmission in forward flow.
59	`Fwd Avg Bulk Rate`	Average bulk transmission rate in the forward direction.
60	`Bwd Avg Bytes/Bulk`	Average number of bytes transmitted per bulk transmission in backward flow.
61	`Bwd Avg Packets/Bulk`	Average number of packets per bulk transmission in backward flow.
62	`Bwd Avg Bulk Rate`	Average bulk transmission rate in the backward direction.
63	`Subflow Fwd Packets`	Average number of packets transmitted in a forward subflow.
64	`Subflow Fwd Bytes`	Average number of bytes transmitted in a forward subflow.
65	`Subflow Bwd Packets`	Average number of packets transmitted in a backward subflow.
66	`Subflow Bwd Bytes`	Average number of bytes transmitted in a backward subflow.
67	`Init_Win_bytes_forward`	Initial TCP window buffer size (in bytes) advertised by client (forward).
68	`Init_Win_bytes_backward`	Initial TCP window buffer size (in bytes) advertised by server (backward).
69	`act_data_pkt_fwd`	Number of forward packets carrying at least 1 byte of actual TCP data payload.
70	`min_seg_size_forward`	Minimum segment size observed in the forward direction.
71	`Active Mean`	Mean duration (in µs) the flow remained active before going into idle state.
72	`Active Std`	Standard deviation of active duration times across the flow.
73	`Active Max`	Maximum duration (in µs) the flow was observed in continuous active state.
74	`Active Min`	Minimum duration (in µs) the flow was observed in continuous active state.
75	`Idle Mean`	Mean duration (in µs) the flow remained idle before transmitting new data.
76	`Idle Std`	Standard deviation of idle periods across the flow.
77	`Idle Max`	Maximum idle time (in µs) observed between active flow transmissions.

#	Feature Name (CICFlowMeter Attribute)	Technical Meaning & Network Security Context
78	`Idle Min`	Minimum idle time (in µs) observed between active flow transmissions.